

ParOSol-X: User Manual

Overview

This manual contains information on the usage of the ParOSol-X web application. This includes:

- usage as a client
- usage as a ParOSol-X operator
- usage as a ParOSol-X administrator

Its very likely you will only be using the site as one of these, so feel free to skip to the relevant section.

This app uses the universities MUSE login service, so any account issues should be brought to the universities technical support.

Requesting Access

When you first visit the website, you will be brought to our homepage. To request access you must already be logged in as a university member, or log in as a university member via the "Sign in" button on the top right of the page.

ParOSol-X

Extreme-scale finite-element analysis services

ParOSol-X is a suite of finite-element analysis (FEA) services that target extreme-scale applications, i.e. where the smallest feature of interest is orders of magnitude smaller than the dimension of the domain.

Examples include: (A) stress analysis of a single bone region, (B) stress analysis in a bone region with variable material representing a population, (C) analysis of contact between trabecular bone and an orthopaedic screw.

Background

All services are based on variants developed starting with the highly-scalable, Cartesian-mesh based, matrix-free FE solver ParOSol (referred to as native ParOSol below). Native ParOSol was developed originally by C. Flaig, P. Arbenz and co-workers, and to our best knowledge, first released c. 2011.

Native ParOSol has been shown to possess excellent weak and strong scalability up to 8000 cores and ~100 billion elements. This underpins the extreme scale capability of the current service. Native ParOSol and our variants have been extensively code-verified as reported [elsewhere](#).

If it is your first time signing in, you will be brought to a page where you can request to access the page as either a client or a ParOSol-X operator. You can also write a comment to communicate with administration during the access request.

Sign up

You have not yet been assigned a role by a system administrator. To ensure you get the role you are looking for, please submit the form below:

Select Role:

- ☐ Client (Submit Jobs)
☐ Operator (Work on Jobs)

Extra Comments:

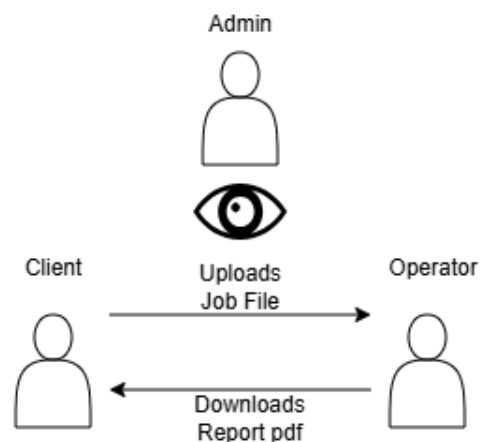
Type here...

Submit

Once administration see your request for access, they will grant you the requested role and you will be able to access the relevant functionalities.

Jobs

Throughout the site and this manual, you will see "jobs" mentioned. Jobs are how files are shared between clients and operators. They contain the clients request(a description of the request, any scan files), an operators report(a pdf written up containing the results of a request) and a status to show the progress of the clients request. A jobs history can be viewed by relevant users, so any changes made to it after creation are visible.



The basic relationship between client, operator, admin and jobs

Client

Creating new jobs

Once signed in, simply press the "Create Job" button found at the top of the screen to start creating a job.

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Fill out the form, and press submit to send the job to our operators. Or, if you don't have all the information and files needed to submit a full request, save the partial submission into your drafts for later revisiting.

the job. input.

Upload progress

Ready

0%

Create Job

Save as Draft

Back

Viewing jobs

To view the jobs list, simply press the "jobs" button on the top bar.

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This will take you to a page showing a list of jobs you have created, including the current status of a job, and any operator who has been assigned to your job. You will be notified by email when changes are made to the jobs status, and can check for further information here.

You can also view any drafts by clicking on the drafts tab, from here you can complete them and send them off as complete jobs.

 University of Sheffield | **INSIGNEO**
Home
Jobs
Create Job
Signed in as acc22rj
Sign out

My Jobs
Drafts

Jobs **1**

All fields
Search...
Search
↓ Date

All Statuses
Showing 1 of 1

TITLE	CLIENT	OPERATOR	STATUS	CREATED AT	
My job	Ryan Jones	-	Pending	17 May, 2026	Details

Downloading job files

Pressing “details” on a job in the job list will allow you to a detailed view of the job. Scrolling through the entry will show buttons to download the PDF file and DICOM file. Once the operator has completed your job and sent a report, the report file will also be available for download.

JOB OVERVIEW

My Job

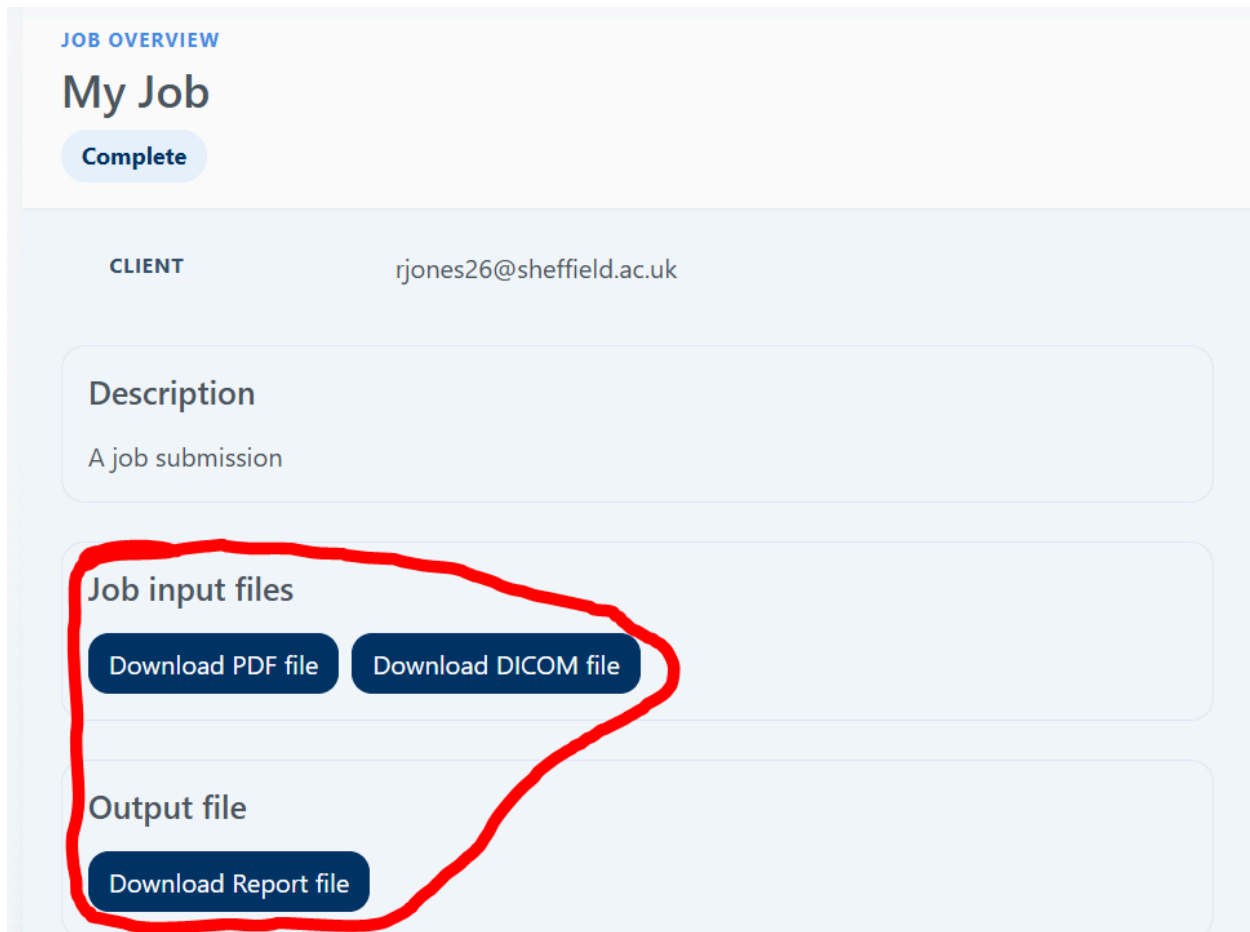
Complete

CLIENT rjones26@sheffield.ac.uk

Description
A job submission

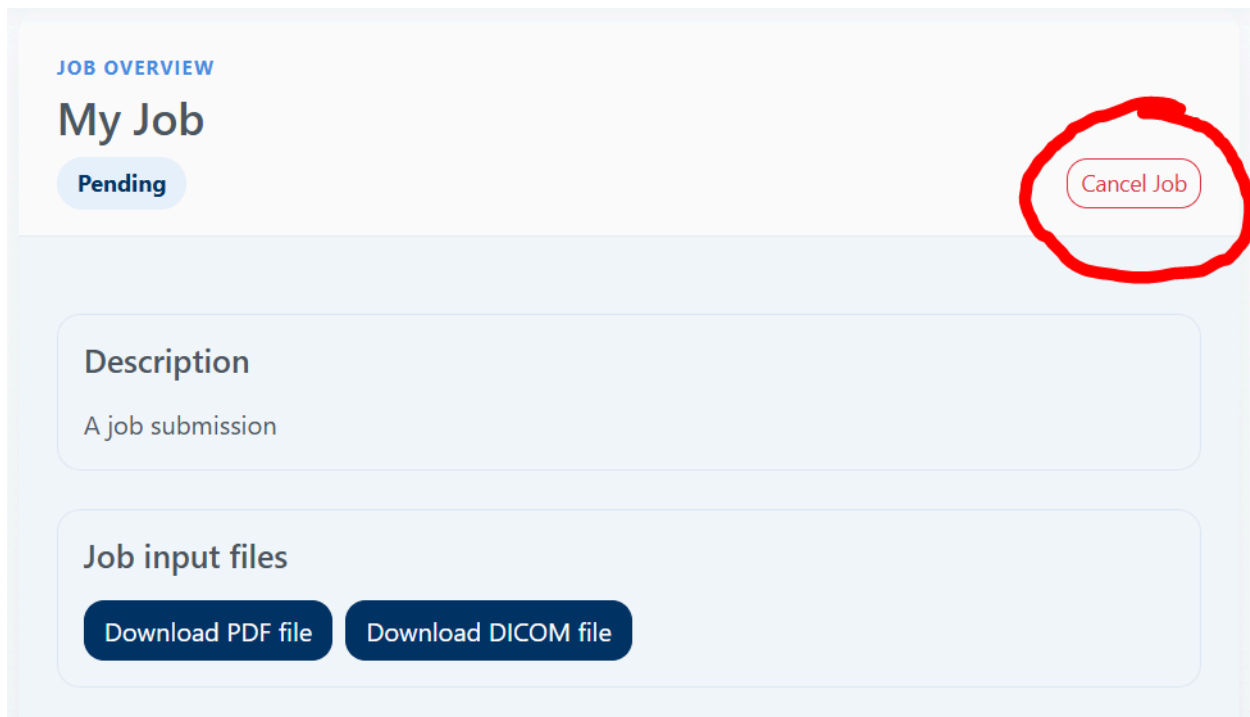
Job input files
[Download PDF file](#) [Download DICOM file](#)

Output file
[Download Report file](#)



Cancelling jobs

To cancel a job, simply press “details” on the job you want to delete from the view jobs table. It will bring you to a detailed view of the job, with a cancel button in the top right. Note that only pending jobs can be cancelled.



Operator

Viewing jobs posted by clients

To view a list of jobs assigned to you and unassigned jobs, press the jobs button on the top bar.

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From here you can switch between viewing jobs assigned to you and jobs without an assigned operator using the tab in the top left, and view the details of a job.

[My jobs](#) [Unassigned jobs](#)

Jobs **1**

All fields ▾

Search...

[Search](#)

[↓ Date](#)

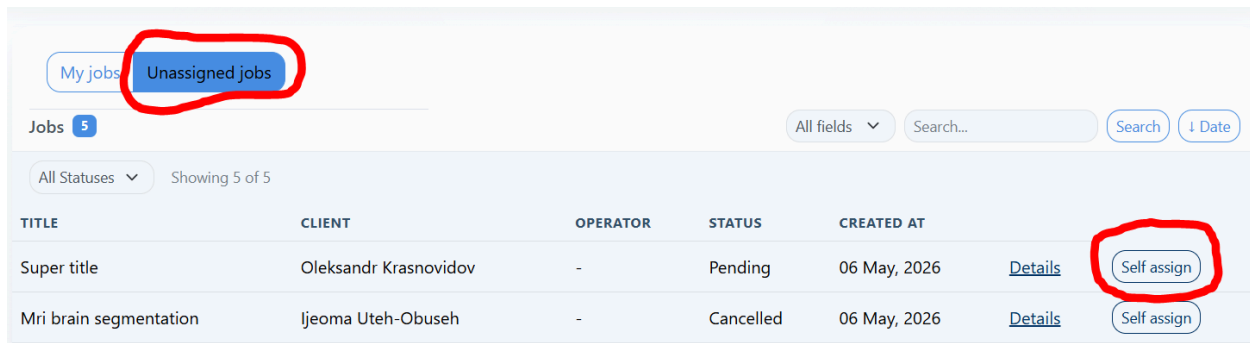
All Statuses ▾

Showing 1 of 1

TITLE	CLIENT	OPERATOR	STATUS	CREATED AT	
My job	Ryan Jones	Ryan Jones	Complete	17 May, 2026	Details

Self Assign Jobs

Jobs are self assigned by operators, to self assign a job navigate to the Unassigned jobs tab of the jobs list, and press the "self assign" button on the job you wish to be assigned.



My jobs		Unassigned jobs			
Jobs 5		All fields ▾ Search... Search ↓ Date			
All Statuses ▾		Showing 5 of 5			
TITLE	CLIENT	OPERATOR	STATUS	CREATED AT	
Super title	Oleksandr Krasnovidov	-	Pending	06 May, 2026	Details Self assign
Mri brain segmentation	Ijeoma Uteh-Obuseh	-	Cancelled	06 May, 2026	Details Self assign

Download Job Files

To download one of your jobs files, navigate to the “My jobs” tab in the jobs list, and press “details” on a job, from here you will be brought to an information screen for the job, containing buttons to download the files and upload your finished report.

University of Sheffield

INSIGNEO

Home

Jobs

Signed in as acc22rj

Sign out

My jobs

Unassigned jobs

Jobs1

All fields ▾

Search...

Search

↓ Date

All Statuses ▾

Showing 1 of 1

TITLE	CLIENT	OPERATOR	STATUS	CREATED AT	
My job	Ryan Jones	Ryan Jones	Complete	17 May, 2026	Details

Description
A job to do

Job input files
[Download PDF file](#) [Download DICOM file](#)

Output file
[Choose File](#) No file chosen
[Upload report file](#)
Report not uploaded yet

Changing job status

You can change a job's status from the details view, this alerts the client to any changes to the job. By default you can assign a job as assigned, in progress or complete, but custom statuses are also available.

JOB OVERVIEW
My Job
In progress [Mark as Complete](#) [Cancel Job](#)

Change status
In progress ▼

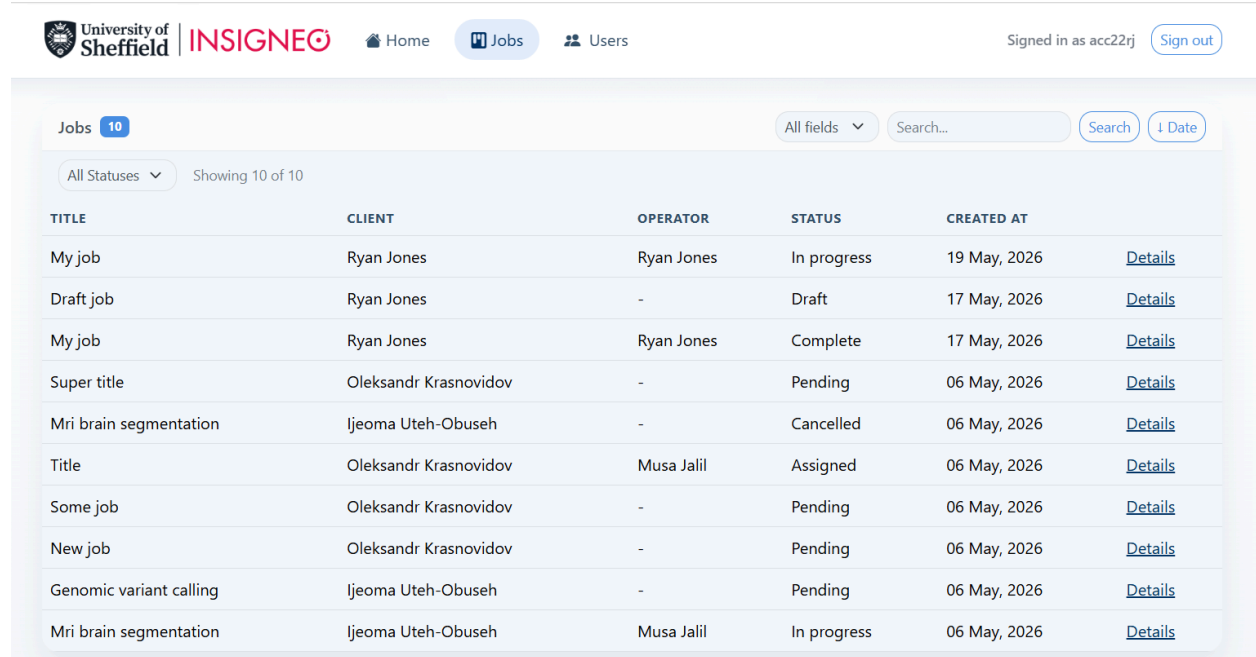
CLIENT [rionec26@sheffield.ac.uk](#)

Jobs can also be cancelled or marked as complete via the buttons in the top right, once a job is marked complete or cancelled, the operator will no longer have access to it without contacting admin.

Admin

Viewing Jobs

Admins can view all jobs by clicking the “jobs” button and edit them by clicking details on any job entry.



The screenshot shows the 'Jobs' management interface. At the top, there's a navigation bar with the University of Sheffield logo, the INSIGNEO logo, and links for Home, Jobs, and Users. The user is signed in as 'acc22rj'. Below the navigation bar, there's a 'Jobs' section with a '10' badge. A dropdown menu shows 'All fields' and a search bar. A table lists 10 jobs with columns for Title, Client, Operator, Status, and Created At. Each row has a 'Details' link.

TITLE	CLIENT	OPERATOR	STATUS	CREATED AT	
My job	Ryan Jones	Ryan Jones	In progress	19 May, 2026	Details
Draft job	Ryan Jones	-	Draft	17 May, 2026	Details
My job	Ryan Jones	Ryan Jones	Complete	17 May, 2026	Details
Super title	Oleksandr Krasnovidov	-	Pending	06 May, 2026	Details
Mri brain segmentation	Ijeoma Uteh-Obuseh	-	Cancelled	06 May, 2026	Details
Title	Oleksandr Krasnovidov	Musa Jalil	Assigned	06 May, 2026	Details
Some job	Oleksandr Krasnovidov	-	Pending	06 May, 2026	Details
New job	Oleksandr Krasnovidov	-	Pending	06 May, 2026	Details
Genomic variant calling	Ijeoma Uteh-Obuseh	-	Pending	06 May, 2026	Details
Mri brain segmentation	Ijeoma Uteh-Obuseh	Musa Jalil	In progress	06 May, 2026	Details

Editing Jobs

Admins can edit the fields of a job from the details job view.

JOB OVERVIEW

My Job

In progress

Mark as Complete

Cancel Job

Change status

In progress

CLIENT

rjones26@sheffield.ac.uk

OPERATOR

rjones26@sheffield.ac.uk

Description

A job to do

Job input files

Delete PDF file

Download PDF file

Delete DICOM file

Download DICOM file

Granting Users Access

Admins can grant users access by pressing the "Users" button on the top bar. From there they can Click the users role to access a dropdown, and assign them the roles "operator" or "client" unlocking the various functionalities associated with either. An Owner account is a special admin account which can appoint other admin.

To remove a user, an admin can change the role to unassigned.

Manage Users

Search users :*

Email or username

Filter by role :*

none

Apply

USERNAME	EMAIL	ROLE
aca24iu	iuteh-obuseh1@sheffield.ac.uk	client
ara23ktd	ktldesilva1@sheffield.ac.uk	client
aca24ok	okrasnovidov1@sheffield.ac.uk	client
aca23ejw	ejwood1@sheffield.ac.uk	operator
aca24maj	majalil1@sheffield.ac.uk	operator